

THEORY AND METHODS

THE USE OF EXPERT OPINION IN THE SEARCH FOR CONSENSUS

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Summary: The problem of two decision-makers who must reach consensus about the probability for the occurrence of an event is considered. The initial probabilities, given by the decision-makers, are updated by information provided by one or more experts. This information can be provided in one of two possible ways by the experts. In the first situation an expert only nominates the event he/she considers the more likely, and in the second situation the expert provides a probability for the occurrence of the event. The decision-makers have an opinion about each expert and this opinion is formed by observation of the expert's performance in the past. It is shown how the decision-makers can decide beforehand, on the basis of this opinion, whether a particular expert will draw their probabilities closer together or not. Thus on the basis of this opinion it can be decided whether to consult an expert or not. The case of two decision-makers can easily be extended to n decision-makers.

1. INTRODUCTION

In this paper two decision-makers, B_1 and B_2 , are considered who must reach consensus about the probability of the occurrence of an event. In their attempt to reach consensus the decision-makers may be able to consult experts and to make use of the information provided by these experts. This information can be made available by the experts in two ways. Consider a parameter space with two points, that is $\Theta = \{\theta_1, \theta_2\}$. In the first case, the

expert only nominates a state which, according to his opinion, is the more likely to be true; and in the second case the expert may provide a probability k for the occurrence of the event $\theta = \theta_1$. The way the information is used by the decision-makers depends on the kind of information that is given by the expert or experts.

Experts are considered as forecasters in almost any situation where repeated predictions must be made. This is also the view of an expert given in DeGroot and Fienberg (1983). In this paper the decision-makers have an opinion about each expert and since repeated predictions of the expert are available, this opinion can be formed by the decision-makers by looking at the expert's performance in the past. An expert is considered as somewhere between the two extreme definitions of experts given in DeGroot (1988). An expert in our situation is not so much a probability assessor, but a person with possibly some special knowledge about the true state of nature. So a "perfect" expert will nominate the true state of nature with probability one. When comparing two experts, one expert shall be considered the better if, in the first case, his probability of nominating the correct state of nature is the higher, or, in the second case, the expectation of the probability that is going to be provided by him is the higher, conditional on $\theta = \theta_1$.

The important issue in this paper is that the decision-makers must reach consensus or come as close as possible to the reaching of consensus. The use of experts does not guarantee consensus, but it will be shown that certain experts will draw the probabilities of the decision-makers closer together. Thus it must be decided which experts to consult and which not to consult. The decision is based on the opinion that is held by the decision-makers about the experts. Thus before consulting an expert the decision-makers can use their opinion of the expert to decide beforehand whether it will be worth their while to consult the expert. This is described for both the cases where the expert nominates a state and where the expert provides a probability for the occurrence of the event $\theta = \theta_1$. It will be shown that the theory for two decision-makers can easily be extended to the case of n decision-makers. Other papers on how to reconcile probability assessments

are those of French (1980 and 1981), Lindley (1983), Lindley, Tversky and Brown (1979), Morris (1977) and Winkler (1986).

2. CONSENSUS IF THE EXPERT NOMINATES A STATE

Consider decision-makers B_1 and B_2 and a parameter space with two points, that is $\Theta = \{\theta_1, \theta_2\}$. According to B_i , $P[T] = p_i$, where T = the event $\theta = \theta_1$.

Since an expert is considered as a forecaster where repeated predictions must be made, the probability of a correct prediction can be estimated from his performances in the past. Let E = the event that "expert says $\theta = \theta_1$ " and

$$(2.1) \quad P[E|T] = P[E^c|T^c] = w.$$

Thus w is the probability that the expert is correct and the expert is also assumed unbiased in the sense that the probability of correct nomination is independent of the true state of nature.

Let $b = \frac{w}{1-w}$, then if $b = 1$ no information is obtained by B_1 and B_2 from the expert. The larger b gets, the higher the opinion of the decision-maker about the expert and the more information is expected from him. A small value of b ($b < 1$) means that the expert can still provide a significant amount of information even though the decision-maker considers him more likely to be wrong than right. However, we can assume, without loss of generality, that $b > 1$. If $b < 1$, it can simply be replaced by $\frac{1}{b}$ and the nominated and non-nominated states interchanged.

The posterior probabilities of B_1 and B_2 can now be calculated. For B_i these probabilities are as follows:

$$(2.2) \quad p'_i = P[T|E] = \frac{bp_i}{bp_i + 1 - p_i} \text{ and}$$

$$(2.3) \quad p_i'' = P[T|E^c] = \frac{p_i}{p_i + b(1-p_i)}, \quad i = 1, 2.$$

Let $p_1 > p_2$ and suppose the expert nominates $\theta = \theta_1$. Consider now the difference in the posterior probabilities given in (2.2). Then

$$(2.4) \quad p_1' - p_2' = \frac{bp_1}{bp_1 + (1-p_1)} - \frac{bp_2}{bp_2 + (1-p_2)} = c_1(p_1 - p_2)$$

where
$$c_1 = \frac{b}{(bp_1 + 1 - p_1)(bp_2 + 1 - p_2)}.$$

Note that $c_1 > 0$ for $1 < b < \infty$, so complete consensus can only be reached ($c_1 = 0$) with the help of a "perfect" expert ($b \rightarrow \infty$). Also, if $p_1 > p_2$, then $p_1' > p_2'$.

The decision-makers will benefit from the expert if $c_1 < 1$, that is, when

$$p_1 p_2 b^2 + (p_1 + p_2 - 2p_1 p_2 - 1)b + 1 + p_1 p_2 - p_1 - p_2 > 0.$$

It can be shown that $c_1 < 1$ for

$$(2.5) \quad p_1 + p_2 < 1; \text{ if } b > \frac{(1-p_1)(1-p_2)}{p_1 p_2} = d,$$

$$p_1 + p_2 \geq 1; \text{ if } b > 1.$$

It is also easy to show that for $p_1 + p_2 < 1$ and a value b' where $d < b < b' < \infty$, then

$$c_1 > c_1' = \frac{b'}{(b'p_1 + 1 - p_1)(b'p_2 + 1 - p_2)}.$$

Similarly for $p_1 + p_2 > 1$, if $1 < b < b' < \infty$ then $c_1 > c_1'$.

The above result simply means that the decision-makers will benefit

more, in terms of convergence of their posterior probabilities, by consulting the "better" expert.

Equivalently the case can be considered where the expert nominates $\theta = \theta_2$ and $p_1 > p_2$.

Then from (2.3)

$$(2.6) \quad p_1'' - p_2'' = c_2(p_1 - p_2)$$

where
$$c_2 = \frac{b}{(p_1 + b(1-p_1))(p_2 + b(1-p_2))} \text{ and again } c_2 > 0.$$

In this case $c_2 < 1$ for

$$(2.7) \quad \begin{aligned} p_1 + p_2 &\leq 1 ; \text{ if } b > 1, \\ p_1 + p_2 &> 1 ; \text{ if } b > \frac{1}{d}. \end{aligned}$$

So the conditions necessary for the decision-makers to benefit from consulting the expert depend on the opinion of the expert. Combining the results (2.5) and (2.7), the posterior probabilities of B_1 and B_2 will be drawn closer together, irrespective of the opinion of the expert, for

$$(2.8) \quad \begin{aligned} p_1 + p_2 &< 1 ; \text{ if } b > d, \\ p_1 + p_2 &> 1 ; \text{ if } b > \frac{1}{d}, \\ p_1 + p_2 &= 1 ; \text{ for all values of } b, b > 1. \end{aligned}$$

If $b = d$ or $b = \frac{1}{d}$ then $p_1' - p_2' = p_1'' - p_2'' = p_1 - p_2$.

Before an expert is consulted, the value of b can be determined for the expert and depending on that value, B_1 and B_2 can decide to consult the expert or not. This process can be repeated for more than one expert until the decision-makers are close enough to consensus in the sense that they can

agree on an action based on some common utility function. The b necessary to satisfy the requirement $p'_1 - p'_2 = c_1(p_1 - p_2) < \varepsilon$ for some $\varepsilon > 0$ can easily be calculated as illustrated in section 4.

It is also possible to consult more than one expert or a group of experts at once and not sequentially as described above. Suppose a group of n experts is consulted by B_1 and B_2 and that k of these experts say $\theta = \theta_1$ with values $b_1, b_2, \dots, b_k$ and $n - k$ say $\theta = \theta_2$ with values $b_{k+1}, b_{k+2}, \dots, b_n$. Then it is easy to show that according to B_i ,

$$(2.9) \quad p_{in} = P[T | \text{group of } n \text{ experts}] = \frac{p_i \prod_{j=1}^k b_j}{p_i \prod_{j=1}^k b_j + (1-p_i) \prod_{j=k+1}^n b_j} \\ = \frac{g_1 p_i}{g_1 p_i + g_2 (1-p_i)}, \quad i = 1, 2,$$

where $g_1 = \prod_{j=1}^k b_j$ and $g_2 = \prod_{j=k+1}^n b_j$.

In this case $p_{1n} - p_{2n} < p_1 - p_2$ for

$$p_1 + p_2 < 1; \text{ if } \frac{g_1}{g_2} < \frac{1}{d} \text{ or } \frac{g_1}{g_2} > d,$$

$$p_1 + p_2 > 1; \text{ if } \frac{g_1}{g_2} < d \text{ or } \frac{g_1}{g_2} > \frac{1}{d},$$

$$p_1 + p_2 = 1; \text{ for all values of } \frac{g_1}{g_2} > 0.$$

3. CONSENSUS IF THE EXPERT PROVIDES A PROBABILITY

In this paragraph the case is considered where the information provided by the expert is a probability k for the occurrence of the event $\theta = \theta_1$. Let $T = \text{event } \theta = \theta_1$, then according to the expert, $P[T] = k$. Suppose

according to B_i ,

$$(3.1) \quad f(k|T) = \frac{\Gamma(r+t)}{\Gamma(r)\Gamma(t)} k^{r-1}(1-k)^{t-1} \text{ and}$$

$$f(k|T^c) = \frac{\Gamma(r+t)}{\Gamma(r)\Gamma(t)} k^{t-1}(1-k)^{r-1}, \quad 0 < k < 1, \quad r > 0, \quad t > 0.$$

Thus $E[k|T] = \frac{r}{r+t}$ and $E[k|T^c] = \frac{t}{t+r}$. So symmetry of the conditional distributions is assumed. The values of r and t can be estimated from the past performances of the expert.

Then according to B_i

$$(3.2) \quad p'_i = P\{T|k\} = \left\{ 1 + \frac{1-p_i}{p_i} \left[\frac{1-k}{k} \right]^{r-t} \right\}^{-1}, \quad i = 1, 2.$$

It is clear from (3.2) that an expert will not be consulted if $r = t$, because then $P\{T|k\} = p_i = P\{T\}$ and no information is obtained by B_i from the expert.

Consider the difference in the posterior probabilities given in (3.2). Then $p'_1 - p'_2 = c_3(p_1 - p_2)$ where $p_1 > p_2$ and

$$c_3 = \frac{\left[\frac{1-k}{k} \right]^{r-t}}{\left[p_1 + \left[\frac{1-k}{k} \right]^{r-t} (1-p_1) \right] \left[p_2 + \left[\frac{1-k}{k} \right]^{r-t} (1-p_2) \right]}$$

(3.3) Note, as in section 2, that $c_3 > 0$, thus if $p_1 > p_2$ then $p'_1 > p'_2$.

We can assume without loss of generality, similarly to section 2, that $r > t$. The objective now is to determine the values of k for which $c_3 < 1$.

$$(3.4) \quad \text{Let } a = \left[1 + \left[\frac{p_1 p_2}{(1-p_1)(1-p_2)} \right]^{(r-t)^{-1}} \right]^{-1},$$

then it can be shown that $c_3 < 1$ for

$$(3.5) \quad \begin{aligned} p_1 + p_2 &< 1 && \text{; if } k < \frac{1}{2} \text{ or } k > a \\ p_1 + p_2 &> 1 && \text{; if } k < a \text{ or } k > \frac{1}{2} \\ p_1 + p_2 &= 1 && \text{; for all values of } k. \end{aligned}$$

The problem in (3.5) is that although r and t can be observed, k is not known to B_1 and B_2 before the expert is consulted. However probabilities that $c_3 < 1$ can be calculated for B_1 and B_2 individually if say, $p_1 + p_2 < 1$. These probabilities can be presented graphically and values for t and r can be found where these probabilities for both decision-makers are greater than a specified value, say α .

Suppose $p_1 + p_2 < 1$. Then the probability that $p'_1 - p'_2 < p_1 - p_2$ according to B_i is

$$(3.6) \quad \begin{aligned} P(B_i) = p_i &\left[\int_0^{\frac{1}{2}} f(k|T) dk + \int_a^1 f(k|T) dk \right] \\ &+ (1 - p_i) \left[\int_0^{\frac{1}{2}} f(k|T^c) dk + \int_a^1 f(k|T^c) dk \right] \end{aligned}$$

If $p_1 + p_2 > 1$ then these probabilities are

$$(3.7) \quad \begin{aligned} P'(B_i) = p_i &\left[\int_0^a f(k|T) dk + \int_{\frac{1}{2}}^1 f(k|T) dk \right] \\ &+ (1 - p_i) \left[\int_0^a f(k|T^c) dk + \int_{\frac{1}{2}}^1 f(k|T^c) dk \right] \end{aligned}$$

Thus an expert will be consulted for those values of r and t , $r > t$ where

$$(3.8) \quad \begin{aligned} P(B_i) &> \alpha \quad i = 1, 2 \text{ if } p_1 + p_2 < 1 \text{ and} \\ P'(B_i) &> \alpha \quad i = 1, 2 \text{ if } p_1 + p_2 > 1. \end{aligned}$$

This will be illustrated by an example in section 4. The process can be repeated for more than one expert until the decision-makers are close enough to consensus.

In this paragraph and the previous one, only two decision-makers are considered. The theory in these two paragraphs can easily be extended to n decision-makers. Consider decision-makers $B_1, B_2, \dots, B_n$ where $P[\theta = \theta_1] = p_i$ for B_i , $i = 1, 2, \dots, n$. Let $p_1 \leq p_2 \leq \dots \leq p_n$, so that $p_1 = \min\{p_1, p_2, \dots, p_n\}$ and $p_n = \max\{p_1, p_2, \dots, p_n\}$. Suppose an expert is consulted in either of the two situations described in section 2 or section 3. Then $B_1, B_2, \dots, B_n$ will each update their probabilities according to formulae (2.2), (2.3) or (3.2). Let $p'_1, p'_2, \dots, p'_n$ denote these updated probabilities then, according to (2.4), (2.6) or (3.3),

$$p'_1 = \min\{p'_1, p'_2, \dots, p'_n\} \quad \text{and}$$

$$p'_n = \max\{p'_1, p'_2, \dots, p'_n\}.$$

Thus the probability given by B_1 will always remain the minimum value even if it is updated by the information provided by the expert. This is also the case for the maximum value given by B_n . Thus if n decision-makers must reach consensus it is, according to this model, only necessary to consider the minimum value and the maximum value, so only the two decision-makers B_1 and B_n are considered.

These models can apply in any situation where two or more decision-makers with common utility function disagree on a course of action due to differences in their subjective probabilities of certain events. Once they accept the qualifications of an expert it can be determined whether consulting the expert would be of any benefit to them.

4. EXAMPLE

Consider the following artificial example to illustrate the above ideas. Suppose there are three decision-makers B_1, B_2 and B_3 that must reach

consensus on $P[T]$. Let $p_1 = 0.6$, $p_2 = 0.1$ and $p_3 = 0.4$ where $p_i = P[T]$ according to B_i . Suppose also that an expert's record shows that $P[E|T] = 0.75$ for the situation in section 2 where he only nominates a state. Thus $b = 3$. The decision whether to consult the expert or not must now be made. It was pointed out in paragraph 3 that it is only necessary to consider the minimum and maximum values which is in this case $p_1 = 0.6$ and $p_2 = 0.1$. Since $p_1 + p_2 = 0.7 < 1$ it follows from (2.8) that the posterior probabilities will be closer together if $b > d$. In this case $d = 6$. Thus the expert must be used if $b > 6$ (or $b < \frac{1}{6}$). Since $b = 3$, the decision-makers will decide not to consult the expert. If they have decided to use this expert and the expert's opinion is that $\theta = \theta_1$ it follows from (2.2) that $p'_1 = 0.8182$, $p'_2 = 0.25$ and $p'_3 = 0.6667$. Thus the probabilities of the decision-makers would have diverged since $p'_1 - p'_2 = 0.5682 > p_1 - p_2 = 0.5$.

Suppose for a next expert $w = 0.875$ was observed, thus $b = 7$. The decision should be made to consult this expert. If this expert says $\theta = \theta_2$ then it follows from (2.3) that $p''_1 = 0.1765$, $p''_2 = 0.0156$ and $p''_3 = 0.087$. In this case the probabilities converge since $p''_1 - p''_2 = 0.109$. If this is not close enough to consensus, another expert can be consulted and so forth until consensus is reached.

Suppose the decision-makers consider it as consensus if $p'_1 - p'_2 < 0.01$ or $p''_1 - p''_2 < 0.01$. If the first expert the decision-makers consult has a value of $b > 823.66$ (or $b < 0.0073$) and the expert states $\theta = \theta_1$, then $p'_1 - p'_2 < 0.01$.

Consider now the case where the expert provides a probability for the occurrence of $\theta = \theta_1$ with p_1 and p_2 as given before. It follows from (3.6) and (3.8) that an expert will be used if t and r are such that $P(B_i) > \alpha$ for $i = 1, 2$. In the next graph it is shown for which values of t and r an expert will be consulted for $\alpha = 0.8$. The area bounded by the contour lines for decision-maker 1 denotes those values of t and r for which $P(B_1) < 0.8$ and equivalently for decision-maker 2.

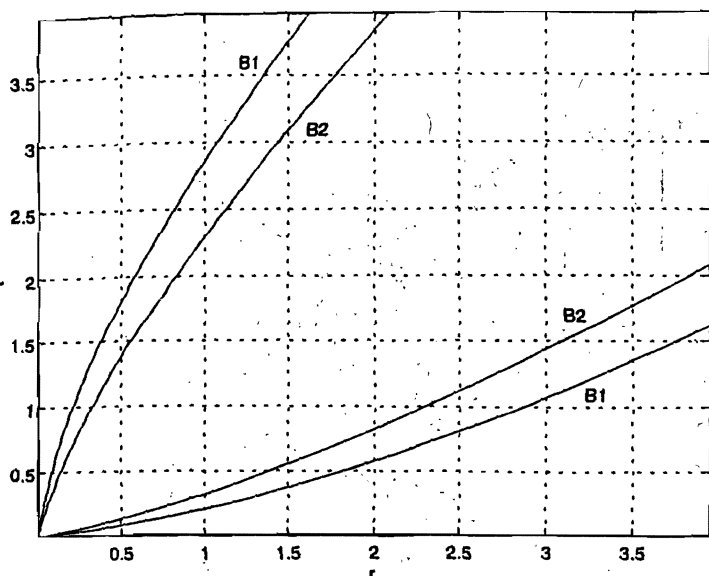


Figure 1: Contour lines of $P(B_1)$ and $P(B_2)$ equal to 0.8 as functions of t and r .

Suppose $r = 2$ and $t = 0.5$ were observed. Then it follows from figure 1 that $P(B_i) > 0.8$, $i = 1, 2$ and that this expert should be consulted. Suppose, according to this expert, $P[T] = k = 0.3$, then from (3.2) $p'_1 = 0.2962$ and $p'_2 = 0.0302$. Since $p'_1 - p'_2 = 0.266$, the probabilities moved closer together.

It must be pointed out that, other than in paragraph 2, there is in this case no guarantee that the probabilities of B_1 and B_2 will converge if t and r are in the regions where $P(B_i) > 0.8$, $i = 1, 2$, as indicated in figure 1. However, once the expert has given his value of k , it must be used since it is information received.

Also, a point like $(r; t) = (2.5; 1)$ might have decision-maker B_2 accepting the expert while B_1 is rejecting him. So to be acceptable to both decision-makers, the expert's parameters should lie in the intersection of the two sets, in this case satisfying B_1 's conditions.

The following graphs show $P(B_1)$ and $P(B_2)$ as functions of t and r .

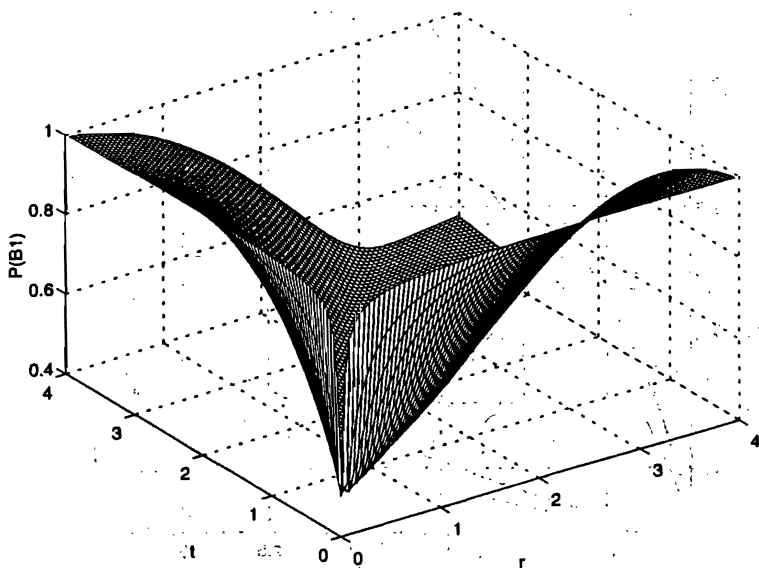


Figure 2: $P(B_1)$ as a function of t and r .

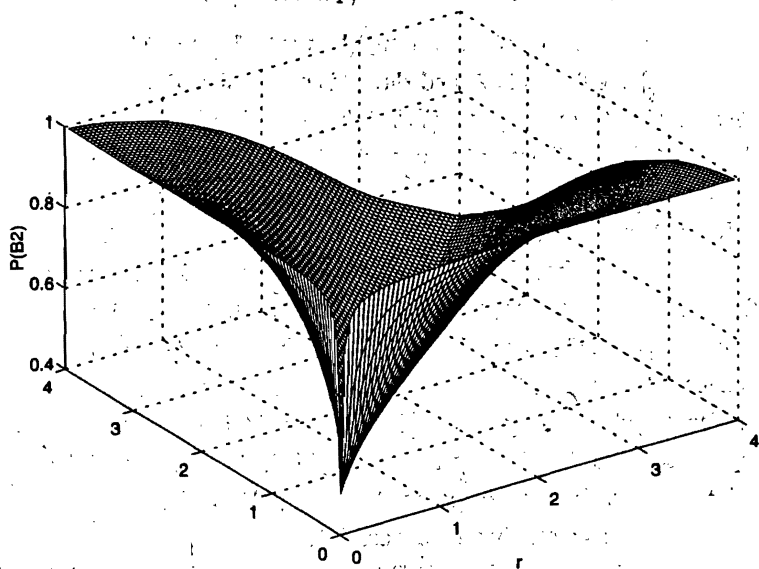


Figure 3: $P(B_2)$ as a function of t and r .

REFERENCES

- DEGROOT, M.H. (1988). A Bayesian view of assessing uncertainty and comparing expert opinion. *Journal of Statistical Planning and Inference*, 20, 295-306.
- DEGROOT, M.H. & FIENBERG, S.E. (1983). The comparison and evaluation of forecasters. *The Statistician*, 32, 12-22.
- FRENCH, S. (1980). Updating of belief in the light of someone-else's opinion. *J. Roy. Statist. Soc., A* 143, 43-48.
- FRENCH, S. (1981). Consensus of opinion. *European Journal of Operational Research*, 7, 332-340.
- LINDLEY, D.V. (1983). Reconciliation of probability distributions. *Operations Research*, 31, 866-879.
- LINDLEY, D.V., TVERSKY, A. & BROWN, R.V. (1979). On the reconciliation of probability assessments. *J. Roy. Statist. Soc., A* 142, 146-180.
- MORRIS, P.A. (1977). Combining expert judgements: A Bayesian approach. *Management Science*, 23, 679-693.
- WINKLER, R.L. (1986). Expert resolution. *Management Science*, 32, 298-303.